Week 7 — Reading Pack

NLP in Pharma Commercial: the regulatory frame for adverse events, what text pharma already mines, interpretability, and the core NLP readings

Applied Data Science and AI in Pharma Commercial · Columbia University · Fall 2026

Read before Monday, October 26 · ~75 minutes · one written reflection (any of B–F) to bring to class

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Sources are limited to regulations and agency guidance, peer-reviewed papers, standard texts, and the IQVIA papers in the course library. Industry passages are quoted verbatim from the IQVIA PDFs in this folder; open-access papers are saved in sources/ (run download_sources.sh); everything is listed with DOIs in Part G. Each section ends with a reflection; write one, bring it.

A. How to read this pack

Tonight's spine is one use case — adverse-event detection in free text — built simply and then opened up so a safety reviewer can see why each note was flagged. The readings supply what the notebook cannot: the rules that define the target (B), the industry's own account of where text comes from and what it is already doing with it (C), the case for explanations in a compliance setting (D), what the literature says social media can and cannot tell you (E), and the core NLP texts (F). Part G lists every source with what it supports.

B. The target, in the regulator's words (objective 3)

Before any model, read the definitions. 21 CFR §314.80(a) defines an adverse drug experience as any adverse event associated with the use of a drug in humans, whether or not considered drug related; serious as death, a life-threatening event, hospitalization, persistent disability, a congenital anomaly or another medically important event; and unexpected as not listed in the current labeling. §314.80(c)(1) requires a 15-day Alert report for experiences that are both serious and unexpected. The FDA's 2001 draft guidance on postmarketing safety reporting adds three things a text model must respect:

> *Applicants should have knowledge of four elements before considering a clinical incident for submission: an identifiable patient; an identifiable reporter; a suspect drug or biological product; an adverse experience or fatal outcome suspected to be due to the suspect drug or biological product.*
>
> — FDA, Postmarketing Safety Reporting for Human Drug and Biological Products Including Vaccines (draft guidance, March 2001), §IV.B — paraphrased list; the four elements are quoted

> *Applicants should review any Internet sites sponsored by them for adverse experience information.*
>
> — FDA (2001), §VI.G, Information on the Internet

> *FDA will consider an applicant responsible for information known to its employees, affiliates, and contractors.*
>
> — FDA (2001), §X

The EU text is parallel: GVP Module VI (Rev 2, 2017), VI.B.1.1.4, says marketing-authorisation holders “should regularly screen internet or digital media under their management or responsibility” for suspected adverse reactions, while digital media outside their control carry an obligation only once the MAH becomes aware of a

case. ICH E2D (2003) gives the international definitions of spontaneous versus solicited reports and the minimum criteria for a valid case.

Why this matters for the model: a saved call note is company knowledge from the moment it exists; the company-sponsored patient community must be screened; a decision to run social listening on an independent forum is a decision to acquire knowledge. And under-reporting is the norm — Hazell & Shakir (Drug Safety 2006) found a median under-reporting rate of 94% across 37 studies — so the cases are in the text and most never leave it.

Reflection B. Take the four elements and write, for each of tonight's three streams (rep notes, forum posts, MI inquiries), which element is hardest to establish from the text and what a model would need to output for a reviewer to decide. Then say which stream you would switch on first and why.

C. What pharma already does with text (objectives 1 & 2)

IQVIA's pharmacovigilance and medical-affairs papers are the industry's own description of where unstructured text comes from and what NLP is already doing with it.

> *PV teams are navigating unprecedented levels of structured and unstructured data, which will continue to increase in complexity and volume from a variety of sources, including clinical trials, medical information calls, insurance claims, social media, and literature.*
>
> — IQVIA, The Future of Pharmacovigilance: Integrating AI with Drug Safety Systems (Hegde & Dosanjh, 2025), The next generation of PV

> *In the background of a medical information call, AI can conduct voice analysis, transcription, and identify adverse events. Sentiment analysis is incorporated to provide greater insight to the safety team when a project is handed off.*
>
> — IQVIA (Hegde & Dosanjh, 2025), The IQVIA approach

> *It is estimated that approximately 80% of Electronic Medical Record (EMR) data exists in unstructured formats such as clinical notes, scanned documents, and free-text entries. These sources often contain rich, nuanced information about treatments, outcomes, and disease progression, details that are rarely captured in structured fields.*
>
> — IQVIA, Harnessing the Power of Artificial Intelligence in Medical Affairs (October 2025), Chart review

> *To streamline the task, IQVIA's AI platform was installed within the organization firewall and domain-specific NLP was used to identify, extract, normalize and collate critical unstructured information from over 34 million clinical notes, with review and validation by the expert-in-the-loop. This resulted in a 100x reduction in manual effort for chart review by site staff, who could instead focus their time on verifying and validating AI derived insights — which were proven to achieve accuracy of 95-99%, including critical endpoints.*
>
> — IQVIA, AI in Medical Affairs (2025), Case study: Streamlining chart review with NLP

> *Automated entity extraction, topic modeling, and sentiment analysis streamline the profiling process by dynamically organizing experts according to therapeutic focus, publication velocity, network importance, and engagement relevance.*
>
> — IQVIA, AI in Medical Affairs (2025), KOL identification and profiling

> *Other types of automation include the automatic transfer of adverse events (AEs) and product complaints (PQCs) to their requisite systems, which helps maintain compliance amid spikes.*

— IQVIA, Harnessing AI to Innovate Medical Information (2025)

> *In medical information, most data is spontaneous and unstructured. That implies a huge responsibility for our confidentiality and how we handle that data, which affects all the other downstream processes, including regulatory and PV.*
>
> — IQVIA, A New, AI-Augmented Horizon for Safety and Pharmacovigilance: Expert Predictions for 2026 (Molloy)

Read the chart-review case as a template for any NLP deployment: inside the firewall, domain-specific model, an expert in the loop, and accuracy reported on the endpoints that matter. Read the Medical Information passages as the operational version of tonight's §9 — intent routing plus an always-on safety detector — and the confidentiality line as the reason the PHI scrub comes before the model.

Reflection C. Pick one commercial text stream the papers do not mention (advisory-board minutes, HCP survey verbatims, call-center chat logs, rep email). Write the use case in the four-column form of the slide: text, method, output, owner — and the regulatory consequence if the model is wrong.

D. Interpretability in a compliance setting (objective 3)

> *The biggest misconception in regulatory is that AI is magical or mysterious. It is neither. The real value of AI, especially in compliance and PV, is augmentation. We are surfacing insights faster and spotting challenges earlier, but we still need governance, transparency, and definition around the context of use. If you cannot explain how your AI came to a decision, it does not belong in a compliance system.*
>
> — IQVIA, Expert Predictions for 2026 (Gyzen)

> *It is difficult to explain how an LLM produces the results it does, why it makes the decisions it makes, or where that information comes from. For most MAHs, establishing trust and confidence in an AI model that lacks full explainability is disconcerting, and though validating a model and enacting controls is a major step towards increasing trust, this inherent risk presents the first hurdle for many.*
>
> — IQVIA, Demanding More from AI Governance in Drug Safety (2025), The challenges

> *The MAH will hold the ultimate responsibility for the AI that is used in their system and will need to document all AI usage information within the PV System Master File (PSMF).*
>
> — IQVIA, AI Governance in Drug Safety (2025), AI governance

> *A human-in-the-loop model absorbs growing data complexity while preserving regulator-trusted decisions. It is not a way station en route to full automation. It is the operating model itself.*
>
> — IQVIA, Right-Sizing Signal Detection: Matching Pharmacovigilance Methodology to Organizational Scale (Raizada, 2025)

Three peer-reviewed anchors. Ribeiro, Singh & Guestrin (KDD 2016) introduced LIME: explain any classifier's prediction locally by perturbing the input, querying the model, and fitting a sparse linear surrogate weighted by proximity — the notebook rebuilds it in fifteen lines. Lundberg & Lee (NeurIPS 2017) unified several attribution methods as Shapley values (SHAP) with consistency guarantees and gave fast exact solutions for linear and tree models. Rudin (Nature Machine Intelligence 2019) argues that for high-stakes decisions we should stop explaining black boxes and use interpretable models instead — explanations of black boxes are approximations, and an approximation that is wrong in the one case that matters is worse than none. For transformers, Jain & Wallace (NAACL 2019) showed that attention weights often do not identify the inputs a model relies on; Sundararajan, Taly & Yan (ICML 2017) give Integrated Gradients, the axiomatic alternative. The FDA / Health Canada / MHRA Good Machine Learning Practice principles (2021) and the FDA's January 2025 draft guidance on AI credibility for regulatory decisions are the governance frame these papers sit inside.

Reflection D. Rudin says prefer interpretable models; a vendor offers you a fine-tuned BioBERT detector with three points more recall and LIME explanations. Write the four questions you would put to the vendor before piloting it, and the one test you would run yourself.

E. What social listening can and cannot find (objective 2)

Sarker et al. (J Biomed Inform 2015) review social-media pharmacovigilance systems: the data sources, the pipeline (collect → classify → extract → normalize), and the open problems — lay language, noise, and whether a poster is an identifiable patient and reporter. Nikfarjam et al. (JAMIA 2015) built ADRMine, extracting ADR mentions from Twitter and DailyStrength with conditional random fields plus word-embedding cluster features — the first widely cited demonstration that embeddings help with lay vocabulary. Golder, Norman & Loke (Br J Clin Pharmacol 2015) reviewed 51 studies and found that adverse events in social media skew to mild, symptom-level events and under-represent serious ones compared with regulatory databases. Freifeld et al. (Drug Safety 2014) compared Twitter mentions with FAERS for 23 products: the rank order of event types correlated, the absolute counts did not. The SMM4H shared tasks (Weissenbacher et al., ACL 2019) are the benchmark where BERT-family models set the state of the art for AE mention classification in tweets.

Read these together with §B: the independent forum is where you find tolerability themes and the patient's own words for them; it is not where you find the pancreatitis case, and looking there creates knowledge you must then act on.

Reflection E. Tonight's notes-trained detector lost recall on forum posts. List three concrete differences between a rep note and a patient post that explain the drop, and for each say whether labels, embeddings, or a rule would fix it.

F. The core NLP readings (objective 1)

Lee et al., BioBERT (Bioinformatics 2020) — the syllabus reading. BERT (Devlin et al., NAACL 2019) further pretrained on PubMed abstracts and PMC full text; consistent gains on biomedical named-entity recognition, relation extraction and question answering. Read §2 (pretraining corpora) and Table 6 (NER) and ask which of tonight's streams looks like PubMed. Alsentzer et al. (Clinical NLP 2019) do the same on MIMIC clinical notes; Gu et al. (ACM TCH 2021) show pretraining from scratch on biomedical text beats continued pretraining and introduce the BLURB benchmark. Reimers & Gurevych (EMNLP 2019) turn BERT into sentence embeddings for search and clustering — the swap-in for the notebook's LSA cells. Blei, Ng & Jordan (JMLR 2003) is the topic-model classic; Grootendorst (2022) is the embedding-based update. Pang, Lee & Vaithyanathan (EMNLP 2002) is the bag-of-words sentiment paper whose NOT_ negation trick the notebook uses; Hutto & Gilbert (ICWSM 2014) is the transparent lexicon approach. Evans, Waller & Davis (2001) and Bate et al. (1998) are the signal-detection statistics in §10.

Reflection F. Compare the pretraining corpora of BERT, BioBERT and ClinicalBERT with the three streams in tonight's data. For each stream, name the model you would try first and the single experiment that would tell you whether the domain model is worth its cost.

G. Every source, and what it supports

Source	What it supports tonight
U.S. Food and Drug Administration. 21 CFR §314.80 — Postmarketing reporting of adverse drug experiences. Code of Federal Regulations (eCFR, current). https://www.ecfr.gov/current/title-21/chapter-I/subchapter-D/part-	Definitions of adverse drug experience, serious and unexpected; 15-day Alert reports for serious and unexpected experiences; periodic reports for the rest.

314/subpart-B/section-314.80	
U.S. Food and Drug Administration. Guidance for Industry: Postmarketing Safety Reporting for Human Drug and Biological Products Including Vaccines (draft). FDA, March 2001. https://www.fda.gov/regulatory-information/search-fda-guidance-documents/postmarketing-safety-reporting-human-drug-and-biological-products-including-vaccines	The four minimum data elements (§IV.B); "applicants should review any Internet sites sponsored by them" (§VI.G); the applicant is responsible for information known to its employees, affiliates and contractors (§X).
International Council for Harmonisation. ICH E2D: Post-Approval Safety Data Management — Definitions and Standards for Expedited Reporting. ICH, November 2003 (Step 4). course library: https://database.ich.org/sites/default/files/E2D_Guideline.pdf	International definitions: spontaneous vs solicited reports; the minimum criteria for a valid case; seriousness; expedited reporting timelines.
European Medicines Agency. Guideline on Good Pharmacovigilance Practices (GVP) Module VI — Collection, management and submission of reports of suspected adverse reactions to medicinal products (Rev 2). EMA/873138/2011 Rev 2, 2017. https://www.ema.europa.eu/en/human-regulatory-overview/post-authorisation/pharmacovigilance-post-authorisation/good-pharmacovigilance-practices-gvp	MAHs "should regularly screen internet or digital media under their management or responsibility" for suspected adverse reactions (VI.B.1.1.4); digital media not under their control carry an obligation only on awareness.
U.S. Department of Health and Human Services. 45 CFR §164.514(b) — De-identification of protected health information (Safe Harbor: the 18 identifiers). Code of Federal Regulations; HHS guidance 2012. https://www.hhs.gov/hipaa/for-professionals/special-topics/de-identification/index.html	Names, dates, medical record numbers and the other identifiers that must be removed before free text can be treated as de-identified.
FDA, Health Canada, MHRA. Good Machine Learning Practice for Medical Device Development: Guiding Principles. October 2021. https://www.fda.gov/medical-devices/software-medical-device-samd/good-machine-learning-practice-medical-device-development-guiding-principles	Ten principles — representative data, independent test sets, performance of the human–AI team, monitoring in deployment — reused here as the governance checklist for an NLP detector.
U.S. Food and Drug Administration. Considerations for the Use of Artificial Intelligence to Support Regulatory Decision-Making for Drug and Biological Products (draft guidance). FDA, January 2025. https://www.fda.gov/regulatory-information/search-fda-guidance-documents/considerations-use-artificial-intelligence-support-regulatory-decision-making-drug-and-biological	The risk-based credibility framework (context of use → model risk → credibility plan) that an AI component in a PV process is expected to follow.
Hazell, L. & Shakir, S. A. W.. Under-reporting of adverse drug reactions: a systematic review. Drug Safety 29(5):385–396, 2006. https://doi.org/10.2165/00002018-200629050-00003	Median under-reporting rate of ADRs across 37 studies: 94% — why mining unstructured text for cases matters.
Evans, S. J. W., Waller, P. C. & Davis, S.. Use of proportional reporting ratios (PRRs) for signal generation from spontaneous adverse drug reaction reports. Pharmacoepidemiology and Drug Safety 10(6):483–486, 2001. https://doi.org/10.1002/pds.677	The PRR and the screening rule PRR ≥ 2, $\chi^2 \geq 4$, n ≥ 3 used in §10.
Bate, A., Lindquist, M., Edwards, I. R., et al.. A Bayesian neural network method for adverse drug reaction signal generation. European Journal of Clinical Pharmacology 54:315–321, 1998. https://doi.org/10.1007/s002280050466	The BCPNN / information component — Bayesian shrinkage for sparse event counts (WHO-UMC).
Harpaz, R., DuMouchel, W., Shah, N. H., et al.. Novel data-mining methodologies for adverse drug event discovery and analysis. Clinical Pharmacology & Therapeutics 91(6):1010–1021, 2012. https://doi.org/10.1038/clpt.2012.50	Review of signal detection across spontaneous reports, EHRs, literature and the web — the map of where AE text comes from.
Karimi, S., Wang, C., Metke-Jimenez, A., Gaire, R. & Paris, C.. Text and Data Mining Techniques in Adverse Drug Reaction Detection. ACM Computing Surveys 47(4):56, 2015. https://doi.org/10.1145/2719920	Survey of lexicon, rule-based and machine-learning approaches to ADR detection in text — the ladder tonight climbs.
Sarker, A., Ginn, R., Nikfarjam, A., et al.. Utilizing social media data for pharmacovigilance: A review. Journal of Biomedical Informatics 54:202–212, 2015. https://doi.org/10.1016/j.jbi.2015.02.004	What social-media pharmacovigilance systems do, their data sources, and the open problems (lay language, noise, identifiability).
Nikfarjam, A., Sarker, A., O'Connor, K., Ginn, R. & Gonzalez, G.. Pharmacovigilance from social media: mining adverse drug reaction mentions using sequence labeling with word embedding cluster features. Journal of the American Medical Informatics Association 22(3):671–681, 2015. https://doi.org/10.1093/jamia/ocu041	ADRMine: ADR extraction from Twitter and DailyStrength with CRFs plus word-embedding features — the first widely cited embedding-based PV system.
Golder, S., Norman, G. & Loke, Y. K.. Systematic review on the prevalence, frequency and comparative value of adverse events data in social media. British Journal of Clinical Pharmacology 80(4):878–888, 2015. https://doi.org/10.1111/bcp.12746	AEs in social media skew to mild, symptom-level events and under-represent serious ones compared with regulatory data — what social listening can and cannot find.
Freifeld, C. C., Brownstein, J. S., Menone, C. M., et al.. Digital drug safety surveillance: monitoring pharmaceutical products in Twitter. Drug Safety 37(5):343–350, 2014. https://doi.org/10.1007/s40264-014-0155-x	Twitter AE mentions vs FAERS: the rank-order of event types correlated, absolute counts did not — reporting-rate caveats.

Weissenbacher, D., Sarker, A., Magge, A., et al.. Overview of the Fourth Social Media Mining for Health (SMM4H) Shared Tasks at ACL 2019. Proceedings of the 4th SMM4H Workshop, ACL 2019. https://doi.org/10.18653/v1/W19-3203	Benchmarks for AE mention classification and extraction in tweets; where BERT-family models set the state of the art.
Devlin, J., Chang, M.-W., Lee, K. & Toutanova, K.. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. NAACL-HLT 2019. https://doi.org/10.18653/v1/N19-1423	The pretrained-transformer recipe that domain models build on.
Lee, J., Yoon, W., Kim, S., et al.. BioBERT: a pre-trained biomedical language representation model for biomedical text mining. Bioinformatics 36(4):1234–1240, 2020. https://doi.org/10.1093/bioinformatics/btz682	Syllabus reading. BERT further pretrained on PubMed abstracts and PMC articles; gains on biomedical NER, relation extraction and QA.
Alsentzer, E., Murphy, J., Boag, W., et al.. Publicly Available Clinical BERT Embeddings. Proceedings of the 2nd Clinical NLP Workshop, NAACL 2019. https://doi.org/10.18653/v1/W19-1909	BERT/BioBERT further pretrained on MIMIC clinical notes — the model for clinician-written text.
Gu, Y., Tinn, R., Cheng, H., et al.. Domain-Specific Language Model Pretraining for Biomedical Natural Language Processing. ACM Transactions on Computing for Healthcare 3(1):1–23, 2021. https://doi.org/10.1145/3458754	Pretraining from scratch on biomedical text beats continued pretraining — and the BLURB benchmark for comparing domain models.
Reimers, N. & Gurevych, I.. Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks. EMNLP-IJCNLP 2019. https://doi.org/10.18653/v1/D19-1410	The sentence-embedding models behind semantic search and BERTopic; the swap-in for §11.
Deerwester, S., Dumais, S. T., Furnas, G. W., Landauer, T. K. & Harshman, R.. Indexing by Latent Semantic Analysis. Journal of the American Society for Information Science 41(6):391–407, 1990. https://doi.org/10.1002/(SICI)1097-4571(199009)41:6<391::AID-ASI1>3.0.CO;2-9	Truncated SVD on the term–document matrix — the offline stand-in for embeddings used in §11.
Blei, D. M., Ng, A. Y. & Jordan, M. I.. Latent Dirichlet Allocation. Journal of Machine Learning Research 3:993–1022, 2003. https://www.jmlr.org/papers/v3/blei03a.html	The probabilistic topic model; NMF (Lee & Seung 1999) is the matrix-factorization alternative used in §7.
Lee, D. D. & Seung, H. S.. Learning the parts of objects by non-negative matrix factorization. Nature 401:788–791, 1999. https://doi.org/10.1038/44565	Non-negative matrix factorization — the topic model used tonight.
Grootendorst, M.. BERTopic: Neural topic modeling with a class-based TF-IDF procedure. arXiv preprint 2203.05794, 2022. https://arxiv.org/abs/2203.05794	Embedding-based topic modeling (sentence embeddings → UMAP → HDBSCAN → c-TF-IDF). Preprint; widely used.
Hutto, C. J. & Gilbert, E.. VADER: A Parsimonious Rule-based Model for Sentiment Analysis of Social Media Text. Proceedings of ICWSM-14 (AAAI), 2014. https://ojs.aaai.org/index.php/ICWSM/article/view/14550	The lexicon-and-rules sentiment approach — transparent, no training data; the model behind the word lists in §7.
Pang, B., Lee, L. & Vaithyanathan, S.. Thumbs up? Sentiment Classification using Machine Learning Techniques. EMNLP 2002. https://doi.org/10.3115/1118693.1118704	Bag-of-words sentiment with the NOT_ negation-scope trick used in the normalizer.
Ribeiro, M. T., Singh, S. & Guestrin, C.. "Why Should I Trust You?": Explaining the Predictions of Any Classifier. KDD 2016. https://doi.org/10.1145/2939672.2939778	Local, model-agnostic explanations by perturbation and a local linear surrogate — rebuilt by hand in §5c.
Lundberg, S. M. & Lee, S.-I.. A Unified Approach to Interpreting Model Predictions. Advances in Neural Information Processing Systems 30 (NeurIPS 2017). https://papers.nips.cc/paper/7062-a-unified-approach-to-interpreting-model-predictions	Shapley-value attributions with consistency guarantees; the shap library used in §5d.
Sundararajan, M., Taly, A. & Yan, Q.. Axiomatic Attribution for Deep Networks. ICML 2017. https://proceedings.mlr.press/v70/sundararajan17a.html	Integrated Gradients — the attribution method of choice for transformer classifiers.
Jain, S. & Wallace, B. C.. Attention is not Explanation. NAACL-HLT 2019. https://doi.org/10.18653/v1/N19-1357	Attention weights often do not identify the inputs a model relies on — the caution on reading attention maps as explanations.
Wiegreffe, S. & Pinter, Y.. Attention is not not Explanation. EMNLP-IJCNLP 2019. https://doi.org/10.18653/v1/D19-1002	The reply: attention can be a useful explanation under the right tests. Optional.
Rudin, C.. Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. Nature Machine Intelligence 1:206–215, 2019. https://doi.org/10.1038/s42256-019-0048-x	Prefer interpretable-by-construction models for high-stakes decisions — the argument for tonight's linear detector.
Doshi-Velez, F. & Kim, B.. Towards A Rigorous Science of Interpretable Machine Learning. arXiv preprint 1702.08608, 2017. https://arxiv.org/abs/1702.08608	What "interpretable" should mean and how to evaluate it (application-grounded, human-grounded, functionally-grounded). Preprint; widely cited.
Molnar, C.. Interpretable Machine Learning: A Guide for Making Black Box Models Explainable (2nd ed.). Self-published, 2022 (free online). https://christophm.github.io/interpretable-ml-book/	The textbook: linear-model interpretation, LIME, SHAP, counterfactuals.
Singhal, K., Azizi, S., Tu, T., et al.. Large language models encode clinical knowledge. Nature 620:172–180, 2023. https://doi.org/10.1038/s41586-023-	Med-PaLM — named in the syllabus; the evaluation framework (factuality, harm, bias) matters more than the

06291-2	model for the GenAI session.
Hegde, A. & Dosanjh, U. (IQVIA). The Future of Pharmacovigilance: Integrating AI with Drug Safety Systems. IQVIA white paper, 2025 — course library. course library: the-future-of-pharmacovigilance-integrating-ai-wp-for-web.pdf	Sources of PV text (clinical trials, medical information calls, claims, social media, literature); AI in the background of MI calls; human-in-the-loop; EDSTP.
Raizada, M. (IQVIA). Right-Sizing Signal Detection: Matching Pharmacovigilance Methodology to Organizational Scale. IQVIA white paper, 2025 — course library. course library: matching-pharmacovigilance-methodology-to-organizational-scale.pdf	"A human-in-the-loop model … is not a way station en route to full automation. It is the operating model itself."
IQVIA. Demanding More from AI Governance in Drug Safety. IQVIA white paper, 2025 — course library. course library: ai-governance-in-safety-whitepaper.pdf	Explainability as the first hurdle for LLMs in PV; the MAH documents AI usage in the PV System Master File; "in the loop or in command".
IQVIA. A New, AI-Augmented Horizon for Safety and Pharmacovigilance: Expert Predictions for 2026. IQVIA white paper, 2025 — course library. course library: ai_for_safety_and_pharma_expert_predictions_for_2026_for-web.pdf	"If you cannot explain how your AI came to a decision, it does not belong in a compliance system." Medical information data is "spontaneous and unstructured".
IQVIA. Harnessing AI to Innovate Medical Information. IQVIA white paper, 2025 — course library. course library: harnessing_ai_to_innovate_mi_wp_for-web.pdf	Contact-center volumes, call transcription, real-time sentiment, automatic transfer of AEs and product complaints to their systems.
IQVIA. Harnessing the Power of Artificial Intelligence in Medical Affairs. IQVIA white paper, October 2025 — course library. course library: bcs2025-3423-10oct---harnessing-the-power-of-ai-in-medical-affairs-white-paper---dale-v2a.pdf	~80% of EMR data is unstructured; NLP chart review of 34 million notes; literature screening reduced by 87%; KOL profiling with entity extraction, topic modeling and sentiment.
IQVIA. Where to Deploy Generative AI Solutions in Life Sciences (commercial). IQVIA white paper, 2025 — course library. course library: where-to-deploy-generative-ai-solutions-in-life-sciences.pdf	"More than 70% of GenAI experiments fail to make it to production"; GenAI excels at text analysis; automate human-like tasks.